Dialogues with AI Reduce Beliefs in Misinformation but Build No Lasting Discernment Skills

Anku Rani*
Massachusetts Institute of Technology
Cambridge, Massachusetts, USA
ankurani@mit.edu

Valdemar Danry*
MIT Media Lab
MIT
CAMBRIDGE, Massachusetts, USA
vdanry@mit.edu

Paul Pu Liang
Massachusetts Institute of Technology
Cambridge, Massachusetts, USA
ppliang@mit.edu

Andrew Lippman
Media Lab
Massachusetts Institute of Technology
Cambridge, Massachusetts, USA
lip@mit.edu

Pattie Maes
Media Lab
MIT
Cambridge, Massachusetts, USA
pattie@media.mit.edu

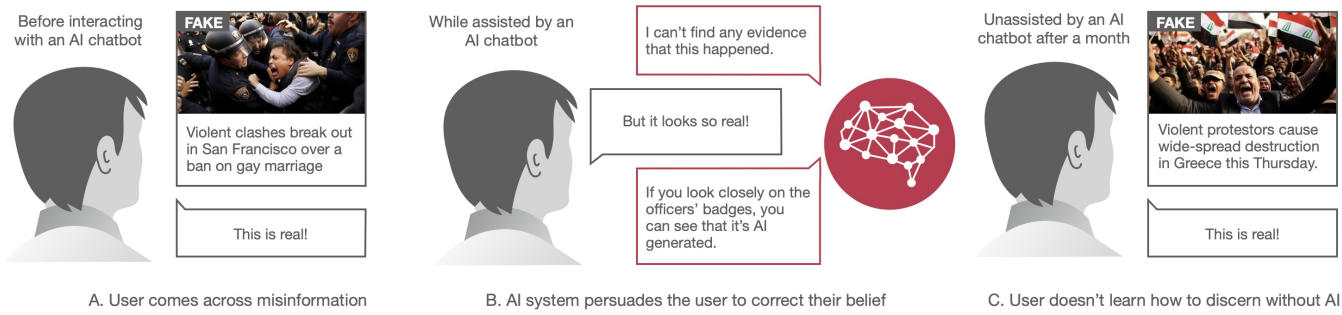


Figure 1: Study overview showing the dependency paradox (where AI assistance provides immediate benefits but undermines long-term capabilities to distinguish real and fake news items without AI assistance). Participants evaluated news items before AI interaction (A), received AI assistance containing evidence with a persuasive response from AI (B), and then evaluated new news items without AI (C). While AI assistance improved immediate performance (21%), misinformation detection skills without AI assistance on new news items declined over time (15.3%), revealing this concerning effect.

Abstract

Given the growing prevalence of fake information, including increasingly realistic AI-generated news, there is an urgent need to train people to better evaluate and detect misinformation. While interactions with AI have been shown to durably reduce people's beliefs in false information, it is unclear whether these interactions also teach people the skills to discern false information themselves. We conducted a month-long study where 67 participants classified news headline-image pairs as real or fake, discussed their assessments with an AI system, followed by an unassisted evaluation of unseen news items to measure accuracy before, during, and after AI assistance. While AI assistance produced immediate improvements during AI-assisted sessions (+21% average), participants' unassisted performance on new items declined significantly by 15.3% in week 4 compared to week 0. These results indicate that while AI may help immediately, it may ultimately degrade long-term misinformation detection abilities.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **HCI design and evaluation methods**; **Empirical studies in interaction design**.

Keywords

Misinformation, Artificial Intelligence, AI, Fake news, Learning, Education, Human-AI Interaction, Overreliance, AI-generated images, Chatbots, Conversational AI, Persuasion, Dialogues, Longitudinal study, Large Language Models, Multimodal Large Language Models, Cognitive Offloading

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*Equal contribution

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1 Introduction

AI Assistants such as ChatGPT, Claude, and Grok are increasingly used to evaluate the credibility of online information, from judging the authenticity of news headlines and viral images to answering whether medical claims [68] or political rumors [45] are true. While recent research suggests such systems can reduce belief in specific false claims [15], it remains unclear whether these conversations teach humans to detect misinformation or merely shift beliefs about false information with AI assistance.

AI-generated misinformation is spreading rapidly online, often gaining widespread traction and causing significant societal harm. For example, AI-generated reports about an explosion near the Pentagon in May 2023 garnered thousands of shares before being debunked (See Appendix section A, Figure 10). This is mostly due to the cognitive biases making people susceptible to false but compelling narratives [59].

While both human and machine-based fact-checking methods [12, 24] have been used to annotate and remove false information from online spheres, they struggle with the speed at which misinformation is produced and spread, or fail to be accurate. Additionally, machine learning-based detection algorithms [67, 78, 79] are computationally expensive and struggle to keep up with the speed at which misinformation can be generated from generative models. Humans are poor at detecting AI-generated content due to the increasing sophistication of generative models and a lack of knowledge about their outputs, such as inconsistent lighting or unnatural textures [5]. Adding to this challenge, research shows that even when false information is explicitly corrected with human or machine-based corrections [12, 24, 67, 78, 79], people often retain their initial beliefs—a persistent cognitive bias known as the continual influence effect [40].

While fact-checking systems have shown mixed results on helping people detect misinformation and correct their beliefs about it [13, 76], recent work on AI chatbots has demonstrated that dialogues with an AI chatbot can reduce belief in conspiracy theories by 22% [15], suggesting potential for AI-assisted belief correction. Furthermore, AI chatbots for education have shown promise in various domains [9–11, 65], particularly when acting as collaborative partners rather than authoritative sources. This makes the AI chatbot a potential tool to teach people how to detect and update their beliefs about misinformation. Yet emerging research on Human-AI interaction suggests that AI chatbots can foster overreliance [20, 44], with users deferring to system outputs even when incorrect [20], potentially undermining the development of independent reasoning skills [7, 27]. This raises an important question: **Do dialogues with AI chatbots actually build lasting misinformation detection skills, or do they create dependency that undermines humans' independent judgment?**

We investigate this question by designing an AI chatbot prompted to generate persuasive dialogues for misinformation detection and by conducting a longitudinal study measuring users' ability to detect misinformation both with and without AI assistance. We summarize our contributions as follows:

(1) **System Design**: We design an AI chatbot that uses persuasive dialogue to help people evaluate news authenticity and detect misinformation. Figure 2 illustrates our system's user interface.
(2) **Longitudinal Study**: We conduct the first month-long, three-phase longitudinal study measuring misinformation detection effects both with and without AI assistance (before and after AI use) (N=67).
(3) **Dependency/Learning Effects**: We demonstrate a trade-off where AI provides immediate accuracy gains (+21%) but leads to a significant decline in unassisted performance after AI interaction (15.3% by week 4 compared to week 0), particularly for detecting fake content; and no significant difference in learning outcomes (unassisted accuracy before AI interaction)
(4) **Conversation Analysis**: We perform a comprehensive analysis of the recorded conversations to automatically classify 24 conversational parameters, examining Human-AI interaction behaviors, providing insights into their correlation with learning outcomes versus dependency development.
(5) **Open Source System and Dataset**: We will publicly release our AI chatbot, 7,203 Human-AI conversation pairs, and 4,536 participant judgments on news authenticity (Real or Fake and a confidence rating) to enable future research.

2 Related Work

2.1 Why Misinformation Persists

The proliferation of AI-generated news media content presents major threats to the reliability of online information ecosystems [25, 80]. As AI generation tools become more sophisticated and accessible, the creation and dissemination of manipulated images alongside misleading headlines has increased dramatically across social media platforms [28]. The combination of visual and textual misinformation content is particularly persuasive, as research has shown that humans tend to place heightened trust in visual evidence while often lacking the skills to detect subtle manipulations or critically evaluate accompanying headlines [39, 58, 75].

Decades of psychological research explain why misinformation is so resistant to correction once it is initially believed. Classic work on rumor and memory demonstrated that false claims continue to shape reasoning even after being explicitly retracted [2, 41], a phenomenon now termed the *continued influence effect*. Meta-analyses and reviews confirm that even detailed corrections rarely eliminate lingering effects after the first exposure to false information [23, 24, 47, 77].

Another well-documented mechanism is repetition of information which increases processing fluency and thus perceived accuracy (also known as the *illusory truth effect*) [26, 35, 55]. Importantly, even prior knowledge does not reliably inoculate against this effect — people with strong factual knowledge are still vulnerable to repeated falsehoods [26]. Recent work shows that reminders and retractions can sometimes backfire, particularly when cognitive load is high or attention is divided [66]. To overcome the continued influence effect, simply correcting false information is insufficient. Instead, research shows that people need coherent alternative explanations that fill the explanatory void left behind to effectively remove the influence of exposure to misinformation [24, 38, 41, 64].

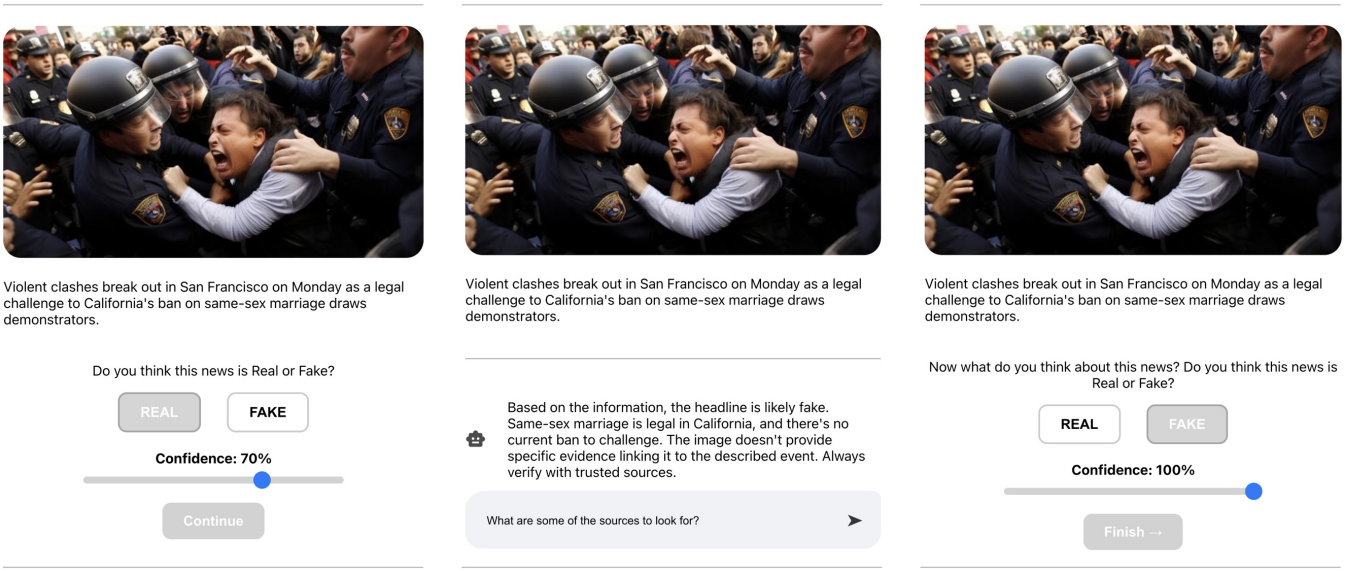


Figure 2: Overview of our AI chatbot's interaction flow. Participants first indicate whether they have seen the news item before, then provide initial authenticity ratings (Step 1). The system then engages them in up to 9 rounds of persuasive dialogue about the item's authenticity (Step 2 shows one example of an exchange). Finally, participants provide updated ratings, allowing us to measure belief change from the AI interaction.

A last relevant mechanism for belief persistence and persuasion in misinformation is the influence of peripheral authority-cues such as professional-looking images or authoritative-seeming sources or explanations [18, 62]. Experiments show that even nonprobative photographs can inflate the perceived truth of trivia statements, and that adding images to news headlines increases perceived accuracy for both true and false items by boosting familiarity and subjective plausibility [53, 69]. Related work on deepfakes and AI-generated visuals finds that realistic, evidential-looking images and videos can substantially increase deception, uncertainty, and trust in misleading political content [32, 34, 73]. In low-attention, high-distraction online environments—where people often lack the motivation or cognitive resources to scrutinize claims—such cues can therefore strongly amplify both susceptibility to and persistence of misinformation.

Together, these findings highlight why misinformation proves so resistant to correction. Heuristics and intuitive reasoning further amplify this vulnerability, as people often rely on fast, effortless judgments rather than deliberate scrutiny, which have been found to overwrite the likelihood of such errors happening. [42, 71, 72]. This persistence underscores the need for interventions that not only correct specific falsehoods but also address the psychological mechanisms that sustain them.

At the same time, psychological evidence indicates that people *can* be trained to think more critically and resist misinformation. Instruction in informal fallacies improves fake news detection [36, 52], while dispositions toward actively open-minded thinking are associated with reduced susceptibility [70]. Encouraging deliberation also helps: prompting individuals to reflect systematically lowers belief in false news headlines [4]. These findings suggest that some resilience against misinformation is achievable.

2.2 Interventions to Reduce Misinformation Beliefs

Extensive research has examined how to mitigate misinformation persistence through corrective information provided before, during, or after exposure. Professional fact-checks and platform warning labels have been found to reduce belief in false stories, but often to a modest effect [12]. Even when corrections are well-designed, people frequently continue to draw inferences from the original misinformation, a phenomenon captured by the continued influence effect [24, 64, 77]. Meta-analyses confirm that while fact-checks do exert a positive influence, their success depends heavily on factors such as timing, repetition, context, and cognitive load during exposure [22, 38, 66].

Recognizing the limitations of post-hoc correction, researchers have explored more proactive strategies. Inoculation or "prebunking" approaches aim to build resilience by warning people about misleading techniques or exposing them to weakened doses of misinformation in advance. This method has shown promise in buffering individuals against subsequent falsehoods [47, 60, 74], but it requires anticipating specific manipulative tactics and often struggles to generalize across the wide variety of misinformation domains.

A complementary line of work targets decision heuristics at the moment of judgment. Subtle prompts that draw attention to accuracy increase discernment both when evaluating and when sharing content online [19, 56, 57, 61]. Similarly, simple accuracy nudges have been found to reduce the spread of false information on social media at scale. Yet these interventions, like corrections and prebunking, typically operate in a one-shot, non-interactive

manner. They provide external inputs but rarely engage with how individuals themselves reason about a claim.

2.3 AI Dialogues and Persuasion

Dialogue-based interventions differ from static fact-checks by responding to the specific reasons people give for their beliefs. In the strongest demonstration to date, Costello et al. asked participants to describe why they endorsed a conspiracy claim and then instructed the AI chatbot to reply persuasively, but respectfully, with counterarguments tailored to those reasons [15]. The result was a large and durable reduction in conspiracy belief. Follow-up analyses clarify the mechanism: the dialogues were effective when the AI chatbot provided *factual counterevidence and alternative explanations* targeted to the user's arguments; when the system was prompted to persuade *without* presenting counterevidence, the effect disappeared [14]. In short, the intervention worked not because an AI was present, but because users experienced having their own arguments acknowledged and then met with "just the facts" that directly undermined those arguments.

Evidence from adjacent AI chatbot system settings is consistent with this evidence-centric account. Studies of AI news chatbots show that perceived trust and persuasiveness increase when the agent confidently presents evidence-backed claims, shaping users' perceptions of sources and narratives [29]. Work on conversational mediation similarly finds that prompt-tuned dialogue strategies can steer how people evaluate arguments during contentious debates [30]. Across these lines of work, the common thread is a *persuasive* conversational stance of the AI chatbots that supplies clear, tailored reasons and sources to move users toward accurate conclusions. While this work shows that conversations with AI chatbots are capable of changing people's beliefs to make them believe misinformation less, it is unclear how much of these effects are caused by reliance on the AI chatbot system or the user internalizing the skills to make good discernments independently.

2.4 Overreliance on AI and the Question of Learning

While persuasive, fact-focused dialogues can correct beliefs in the moment, there is a growing concern that such answer-giving strategies foster reliance on the AI rather than cultivating independent reasoning. Evidence from across HCI and psychology points to this risk. Studies of AI-assisted decision-making repeatedly find that users defer to system outputs even when they are erroneous, a phenomenon termed *overreliance* [7, 63]. Providing explanations or rationales does not consistently mitigate this effect; in some cases, explanations increase deference by making outputs appear more authoritative [63].

Recent work in Human–AI interaction illustrates the same dynamic in high-stakes reasoning contexts. Participants using AI systems that confidently "tell" rather than "ask", report going along with the system because it sounds knowledgeable, thereby reducing their own engagement in critical evaluation [20]. Similar patterns are observed in writing and analysis tasks: generative AI tools homogenize style [1], reduce self-reported cognitive effort [46], and accumulate what has been called *cognitive debt*, where reliance grows and effort declines over repeated use [44]. These observations align with classic automation-bias and algorithm-(aversion/appreciation) results showing systematic deference to automated recommendations under authority cues and cognitive load [21, 48, 54], and with cautionary findings from wearable guidance that directive cues invite default-to-system behavior unless prompts are designed to elicit reflection [19, 20].

Taken together, these findings suggest that the very strategies that make persuasive dialogues effective—providing confident, tailored factual answers—might also be those most likely to induce reliance and discourage the type of cognitive engagement that might lead to learning [7]. In the misinformation domain, this raises a critical open question: do AI dialogues merely persuade users about specific claims in the moment, or do they help users develop misinformation detection skills to evaluate new claims on their own? Our study directly addresses this gap by separating immediate gains during assisted interactions from unaided performance on novel misinformation, testing whether persuasive AI dialogues foster discernment or dependency.

3 System Design

We designed a web-based AI chatbot to investigate whether dialogues with AI can help humans learn to distinguish between real and fake news content over the course of a month-long study. The system presents participants with real and AI-generated news headline-image pairs, then engages them in conversations designed to guide them toward accurate authenticity assessment. In the following, we discuss the system overview, data curation, and system evaluation.

3.1 System Overview

To evaluate the effects of an AI chatbot on misinformation discernment, our system implementation closely follows the state-of-the-art approach [15] for generating persuasive dialogues. Combining LLMs with web search has proven to improve models' accuracy for fact-related queries [3, 51]. We therefore integrated web search for headlines and artifact detection for images, adapting the prompt from Costello et al. [15] to improve our system's accuracy on current misinformation examples in our dataset. The system consists of three main components: (i) React-based front-end for user interface, (ii) OpenAI GPT-4o prompted for persuasive dialogue generation (prompts can be found in Appendix B), and (iii) Google Custom Search API for real-time information retrieval. Figure 3 illustrates the system architecture. The React front-end manages the user interface, presenting news headline-image pairs and collecting participant responses, including initial authenticity ratings and confidence levels (See Figure 2). Those ratings are then sent to an AI chatbot system running GPT-4o, prompted to generate contextually appropriate persuasive responses to persuade users to change their beliefs about the headline-image pair if they were wrong (refer to Figure 14 to view an example from one such conversation). If they were correct, the AI response reinforces the user's belief, and the user can still interact with the AI system (refer to Figure 13 to view an example from one such conversation).

This is implemented through a structured conversational approach where GPT-4o: (1) acknowledges the participant's reasoning, (2) presents relevant factual evidence retrieved from Google Search,

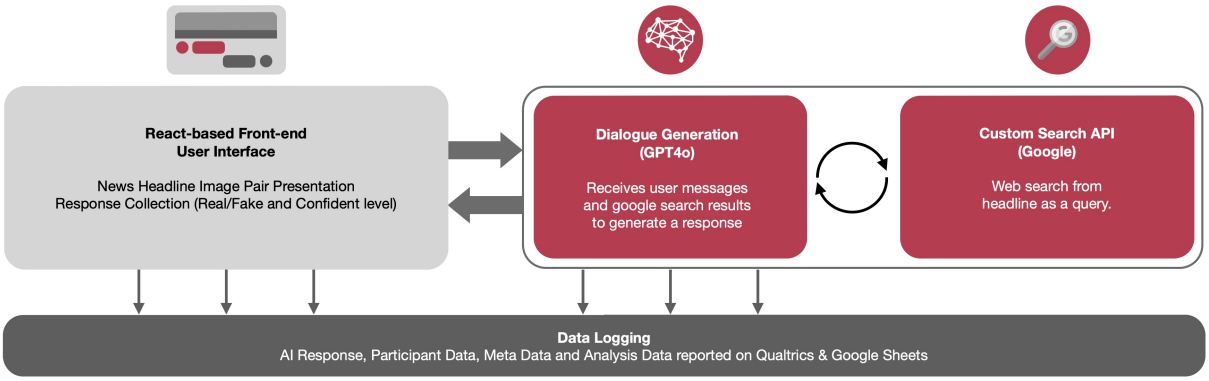


Figure 3: The system integrates three core components: (1) a React-based front-end for presenting news headline-image pairs and collecting participant responses (authenticity ratings and confidence levels), (2) OpenAI GPT-4o for generating contextually appropriate persuasive dialogue based on participant assessments, and (3) Google Custom Search API for real-time searching into the web with headline and fetching relevant information. The architecture supports iterative conversations with comprehensive data logging through Qualtrics and Google Sheets for subsequent analysis. Bidirectional communication between GPT-4o and the search API ensures responses incorporate current factual information, while the feedback loop enables up to 9 rounds of persuasive dialogue per news item.

and (3) guides users through iterative persuasive dialogue toward more accurate assessments of the news item's authenticity. The system architecture supports up to 9 rounds of conversation per news item. The conversation stops when 9 rounds are completed, or when the AI analyzes from the conversation that the user is done talking about the current news item, and then it moves to the next news item. All interactions are logged, including participant responses, AI-generated content, and metadata for subsequent analysis. Data is stored using both Qualtrics integration and Google Sheets API for analysis.

3.2 Data Curation

We randomly selected 55 news items from the MiRAGeNews dataset [37], which comprises 12,500 high-quality image-caption pairs—both real news items and AI-generated pairs produced using state-of-the-art generators—along with their ground truth labels (Real or Fake). Figure 4 shows representative examples from the dataset. These selected news items were processed through the architecture discussed in Figure 3 and prompted to output "Real" or "Fake". The output was then compared with the ground truth labels from the MiRAGeNews dataset. The system misclassified 6 items, which we removed from the dataset. The remaining 49 items formed our final dataset, which was distributed across three study phases. Throughout the three phases, participants evaluated 4 randomly selected news items (2 real and 2 fabricated), followed by confidence ratings on 4 previously unseen items to assess whether they could correctly distinguish between real and fake news in unassisted sessions. In phases 2 and 3, the unassisted evaluation sessions immediately after AI interactions included 6 items: 4 novel unseen items and 2 items previously encountered in the preceding phase. However, we excluded 2 previously seen items from the preceding phase to measure immediate learning on misinformation discernment. The items span diverse topics, including elections, protests, court proceedings, diplomatic events, health matters, and disasters, ensuring broad coverage of typical news categories where misinformation commonly appears.

3.3 System Evaluation

Before deploying the system with participants, we conducted extensive validation of the AI's accuracy to ensure reliable performance. We initially tested our system on a sample of eight items across three different prompt variations using GPT-4o under four different conditions based on available information: (i) Google search on headline with images passed to GPT-4o, (ii) Google search on headline without images, (iii) no Google search with images on GPT-4o, and (iv) no Google search and headline only on GPT-4o. We tested prompts designed for artifact detection only, persuasion/belief change only, and a combined forensic expert and persuasion approach. Figure 11 contains the detailed prompts, and Table 1 contains the evaluation metrics for all conditions.

We measured overall accuracy, rejection rate, and accuracy among non-rejected responses. Overall accuracy is defined as the model's ability to correctly differentiate between real and fake news headlines and their associated images across the four conditions. Rejection rate is defined as the frequency at which the model declines to provide an answer for a given prompt. Non-rejected accuracy is defined as the model's accuracy in cases where it provided a response. The combined prompt with Google search for headlines and image analysis for images passed to GPT-4o (Condition 1) achieved optimal performance with 100% accuracy and 0% rejection rate on the sample, leading to its selection for the system design.

3.4 Interaction Flow & Conversation Data

Each session follows a standardized protocol. Participants first indicate whether they have seen the news item before, then provide an initial authenticity rating using a binary choice (REAL/FAKE) and confidence slider (0-100%). The system then initiates a conversation where the AI attempts to persuade users toward the correct answer



Figure 4: Sample dataset from MirageNews [37] containing news image-headline pairs with ground truth labels. The dataset includes both real news stories (left: aircraft wreckage recovery, right: President Biden) and fabricated content (center: fake World Economic Forum gathering, fake violent clashes), demonstrating the mixed real/fake nature of misinformation datasets used to evaluate AI detection capabilities.

Prompt	Evaluation Metrics	Condition 1 Google Search on Headline + Image on GPT-4o	Condition 2 Google Search on Headline + No Image on GPT-4o	Condition 3 No Google Search + Image on GPT-4o	Condition 4 No Google search + Headline on GPT-4o
Prompt 1: News Forensic Expert (Artifact detection only)	Overall Accuracy	71.43%	78.57%	35.71%	0.00%
	Rejection Rate	21.43%	21.43%	50.00%	92.86%
	Accuracy (Non Rejected)	90.91%	100.00%	71.43%	0.00%
Prompt 2: Persuasion (Persuasive Prompting)	Overall Accuracy	71.40%	78.57%	28.57%	0.00%
	Rejection Rate	14.20%	14.20%	78.57%	92.00%
	Accuracy (Non Rejected)	83.30%	91.60%	66.66%	0.00%
Prompt 3: Combined (News Forensic Expert + Persuasion)	Overall Accuracy	**100%**	78.5%	42.8%	0%
	Rejection Rate	**0%**	14.20%	21.42%	71.4%
	Accuracy (Non Rejected)	**100%**	91.60%	54.45%	0%

Table 1: Comparison of different prompt conditions and their impact on accuracy and rejection rates. These prompts are described in Figure 11. The combined news forensic expert prompt with persuasion achieves optimal performance with 100% accuracy and 0% rejection rate, making it the most effective approach for misinformation detection.

if they were wrong (refer to Figure 14 to view an example from one such conversation). If they were correct, the AI response reinforces the user's belief, and the user can still interact with the AI system (refer to Figure 13 to view an example from one such conversation). Our data shows that users changed their initial assessment in 81.14% of conversations when initially incorrect, and maintained their assessment in 98.68% of conversations when initially correct.

After the dialogue concludes (maximum 9 exchanges), participants provide a final rating for the news item using the same interface. All interactions are logged with timestamps, including initial ratings, conversation turns, and final ratings. The system records participant responses, AI responses, and metadata (participant ID, item details) for subsequent analysis. Data is stored both on Qualtrics and via the Google Sheets API. Appendix Section D provides additional examples and details on the conversation data.

4 Study Design

We conducted a longitudinal study to understand how AI assistance affects people's ability to detect misinformation over time. The study ran for four weeks with three testing sessions at weeks 0, 2, and 4. Each session had three phases: (i) **Before AI**: participants assessed news items independently, (ii) **With AI**: they interacted with our AI chatbot about the same news items, and (iii) **After AI**: they independently evaluated new unseen items without AI assistance. This design enabled us to measure both immediate AI assistance effects and lasting skill development—distinguishing between momentary performance gains and genuine improvement in their independent detection abilities. Figure 5 shows the experimental setup.

4.1 Participants

We recruited 105 participants through Prolific, an online research platform, targeting English-proficient adults aged 18 and older from the US and UK. Of the initial 105 participants, 28 dropped out before completing all three sessions, and we excluded an additional 10 participants who failed more than one attention check per session. Attention checks were implemented to ensure participant engagement throughout the study. This resulted in a final sample of 67 participants who completed the entire longitudinal study and provided reliable data for analysis.

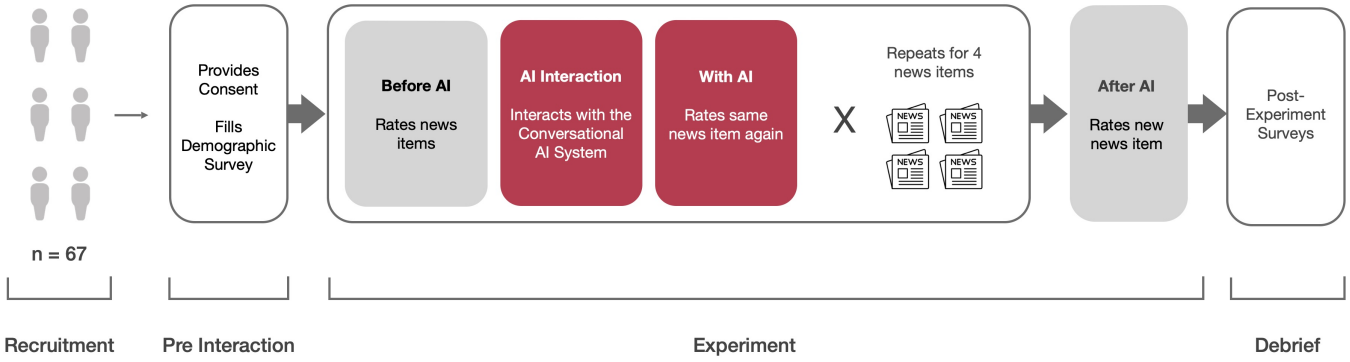


Figure 5: Longitudinal study design workflow showing the three-phase experimental protocol. Participants complete Before AI (rates news item as Real or Fake and provides confidence rating before interacting with AI), AI Interaction (interacts for up to 9 rounds of conversations about the news item), With AI (re-rates the news item after conversation), and After AI (rates new unseen news items) phases across multiple sessions at weeks 0, 2, and 4. This design separates immediate AI assistance effects from independent skill development, enabling measurement of whether participants develop lasting misinformation detection abilities.

The final sample had a mean age of 40 years (range: 20-63) with relatively balanced gender representation (53.7% male, 46.3% female). Participants were predominantly from the United States (76.1%) and the United Kingdom (23.9%). The ethnic composition included White participants (56.7%), Mixed ethnicity (14.9%), Black participants (14.9%), Asian participants (6.0%), Other ethnicities (4.5%), and Unknown ethnicity (3.0%). Figure 12 presented in the Appendix section C displays the complete demographic distribution of the study sample.

Participants received $8 compensation for each 40-minute session and a $3 bonus for completing all three phases. This study was approved by our Institutional Review Board at MIT (Protocol #2505001647), and all participants provided informed consent.

4.2 Experimental Protocol

Each weekly session followed the same three-phase protocol. In the **Before AI** phase, participants provided initial authenticity ratings on 4 randomly selected headline-image pairs (2 real, 2 fake) without any AI assistance. Next, in the **With AI** phase, participants interacted with the AI chatbot on the same 4 headline-image pairs for up to 9 rounds, engaging in dialogue to refine their judgments and potentially change their initial assessments. Finally, in the **After AI** phase, participants evaluated previously unseen items without AI assistance to measure independent skill development. In week 0, participants evaluated 4 new items, while in weeks 2 and 4, they evaluated 6 items: 4 new unseen items and 2 items from the previous week to assess retention. This design allowed us to capture baseline performance, immediate AI assistance effects, and whether participants developed skills for misinformation detection.

4.3 Data Collection

We collected quantitative, qualitative, and exploratory data to assess both performance and participant characteristics. Our primary measures included accuracy, scored as binary classification performance (1 = correct, 0 = incorrect), confidence measured as self-reported certainty on a 0-100% scale, and belief change calculated as the difference between initial and final ratings during AI assistance.

We collected additional measures to characterize our sample and verify data reliability for every participant. For **attention checks**, we implemented a two-stage process: first, participants who failed 2 pre-interaction attention checks were removed from the study, and second, after each AI interaction, participants answered 2 additional attention checks, leading us to exclude 10 participants who failed these post-interaction checks. We measured participants' **prior AI experience** using three questionnaires: (i) Frequency of AI chatbot usage (ChatGPT, Gemini, Claude, Perplexity) from "Never" to "More than 6 hours a day," (ii) Frequency of AI image generation tool usage (ChatGPT, Midjourney, Meta AI) from "Never" to "Everyday," and (iii) AI literacy that contains questions like "I understand the basic concepts of AI" and rated from strongly disagree to strongly agree. We present additional details in Appendix E Figure 16. We also collected qualitative feedback through three open-ended questions about whether participants felt the AI taught them detection skills, their thinking process during AI dialogue, and any technical or other issues they experienced.

5 Analysis

Our dependent variable was item-level accuracy (1 = correct, 0 = incorrect). Analyses were restricted to participants who completed all three phases (before, with, after) at Weeks 0, 2, and 4 and did not fail more than one attention check (n=67). Items flagged as "seen before" were excluded from analysis.

We estimated treatment effects using linear probability models (Ordinary Least Squares regression on accuracy) with participant fixed effects and Heteroskedasticity-Consistent 2 (HC2) robust standard errors, following prior work on AI dialogues and belief updating [15]. This approach yields interpretable percentage-point effects while absorbing stable individual differences in baseline accuracy.

Our main tests focus on two planned within-week contrasts: (i) Δ(with − before), the in-session gain on assisted items when participants conversed with the AI, and (ii) Δ(after − before), the unaided learning to new items following interaction. These contrasts are estimated separately for each week and reported in percentage points with HC2 robust 95% confidence intervals.

To capture changes in unaided performance over time, we report our results on *after*-phase trials only, testing absolute differences in unaided accuracy across weeks (Week 4 vs. Week 0; linear trend). This is done through a linear mixed-effects regression model to assess changes in unassisted accuracy over time, focusing specifically on the "after AI support". The model includes week as a continuous predictor (coded as 0, 2, 4 for the three time points) and participant fixed effects to control for individual differences in baseline accuracy. We use heteroskedasticity-consistent (HC2) robust standard errors to account for potential variance heterogeneity across participants and time points. The linear trend coefficient, β, represents the change in accuracy per two-week interval, allowing us to quantify the rate of decay in unassisted performance. We additionally test the specific contrast between Week 4 and Week 0 using a linear hypothesis test with the same robust covariance matrix to confirm the overall decline from initial to final time points. For descriptive context, we plot unaided accuracy by week with Wilson 95% confidence intervals.Because skill development may differ by veracity, we replicated the after-only analyses separately for fake versus real items. All tests are two-sided with $\alpha = .05$.

To identify which conversational strategies influenced fake news detection performance, we applied LLM-as-a-judge methodology to classify each Human-AI dialogue into 21 binary strategy categories. These categories captured specific conversational moves across six domains: general behaviors (e.g., asking guiding questions, probing deeper), evidence strategies (e.g., image forensics, source checking), reasoning strategies (e.g., logic plausibility checks, alternative explanations), emotional strategies (e.g., bias awareness, emotional trigger detection), knowledge activation (e.g., prior knowledge recall, fact comparison), and metacognitive strategies (e.g., confidence calibration, process reflection). Each classifier used structured prompts with explicit descriptions, examples, and binary classification options (0/1) (See all prompts in Appendix E.2). We applied reliability filtering to ensure robust correlations, including only strategies with occurrence rates between 5% and 95%, at least 10 positive classifications, and sufficient variance (>0.05).

We calculated Pearson correlation coefficients, r, between each conversational strategy (binary coded by LLM-as-a-judge) and accuracy outcomes under two conditions: AI-assisted accuracy (during Human-AI interaction) and unassisted accuracy (after AI interaction). Correlations were computed separately for fake news items to focus on misinformation detection. For each qualifying strategy, we calculated the difference between unassisted and AI-assisted correlations (diff = $r_unassisted$ - $r_assisted$) to identify whether strategies promoted learning transfer (positive difference) or AI dependence (negative difference). We also computed standard deviations from the strategy occurrence distributions to assess variability across conversations.

6 Quantitative Results

6.1 Descriptive Overview

The final sample consisted of 67 participants who completed all three weekly sessions (Weeks 0, 2, and 4), from the initial 105 recruited participants. The analysis included 4,536 total item ratings across the three phases (1,512 ratings per phase). Participants engaged in an average of 9 conversational turns with the AI system per news item, generating 7,203 pairs of Human-AI conversations. Additional details on conversations are present in Appendix section D.

6.2 Unassisted accuracy before AI interactions showed no to modest declines over time, indicating limited skill learning

To assess whether participants developed improved discernment skills from their AI interactions, we examined baseline accuracy before interacting with the AI in each weekly session (before any AI support). This "Before AI Support" measure provides a cleaner indicator of learning transfer than post-interaction performance, as it captures participants' independent abilities before any AI interaction. Before AI, accuracy showed a modest decline over the four-week period. Week 4 baseline accuracy was 6.8 percentage points lower than Week 0, though this difference did not reach statistical significance (95% CI: −15.3 to +1.7 pp, $p = .127$). A linear trend analysis similarly indicated a decline of −3.5 pp per two-week step ($\beta = -0.035$, $SE = 0.022$, $t = -1.55$, $p = .11$), while Week 2 showed a small non-significant increase ($\Delta = +3.4$ pp, $p = .49$). These results suggest that repeated interactions with the AI system did not substantially improve participants' independent discernment abilities between sessions.

6.3 Persuasive AI substantially improved accuracy when assisting the user

In our results, participants showed large short-term gains when interacting with the persuasive AI. Across all weeks, accuracy on assisted items was markedly higher than before assistance, with an average uplift of about +21.3 percentage points (pp) (Week 0: +22.4, Week 2: +15.3 pp, Week 4: +26.1 pp). These week-by-week effects were robust, and the assisted gain was somewhat larger at Week 4 compared to Week 0 ($\Delta = +3.6$ pp). This pattern suggests that the AI chatbot consistently corrected misperceptions during conversations and that the strength of this persuasion effect may have grown over time. See Figure 6.

6.4 People's accuracy with AI did not persist when they were unassisted after AI interactions, and their detection abilities declined

Our results show that after interacting with the AI when it is unavailable, their accuracy on unaided items ("After AI support") declined steadily over the four-week period. When judging new items without AI assistance, accuracy decreased linearly across

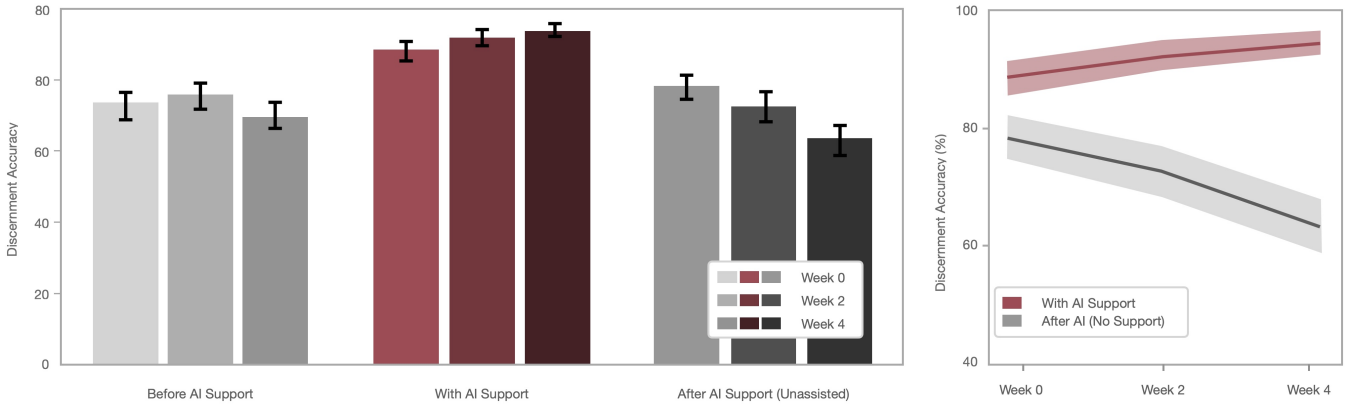


Figure 6: Quantitative results showing participant discernment accuracy of true and false information *before*, *with*, and *after* AI support across the four weeks. Left: Side-by-side comparison of accuracy differences for before AI support, with AI support, and after AI support (unassisted). Right: Line plot showing changes in accuracy over Week 0, Week 2, and Week 4. 95% confidence intervals.

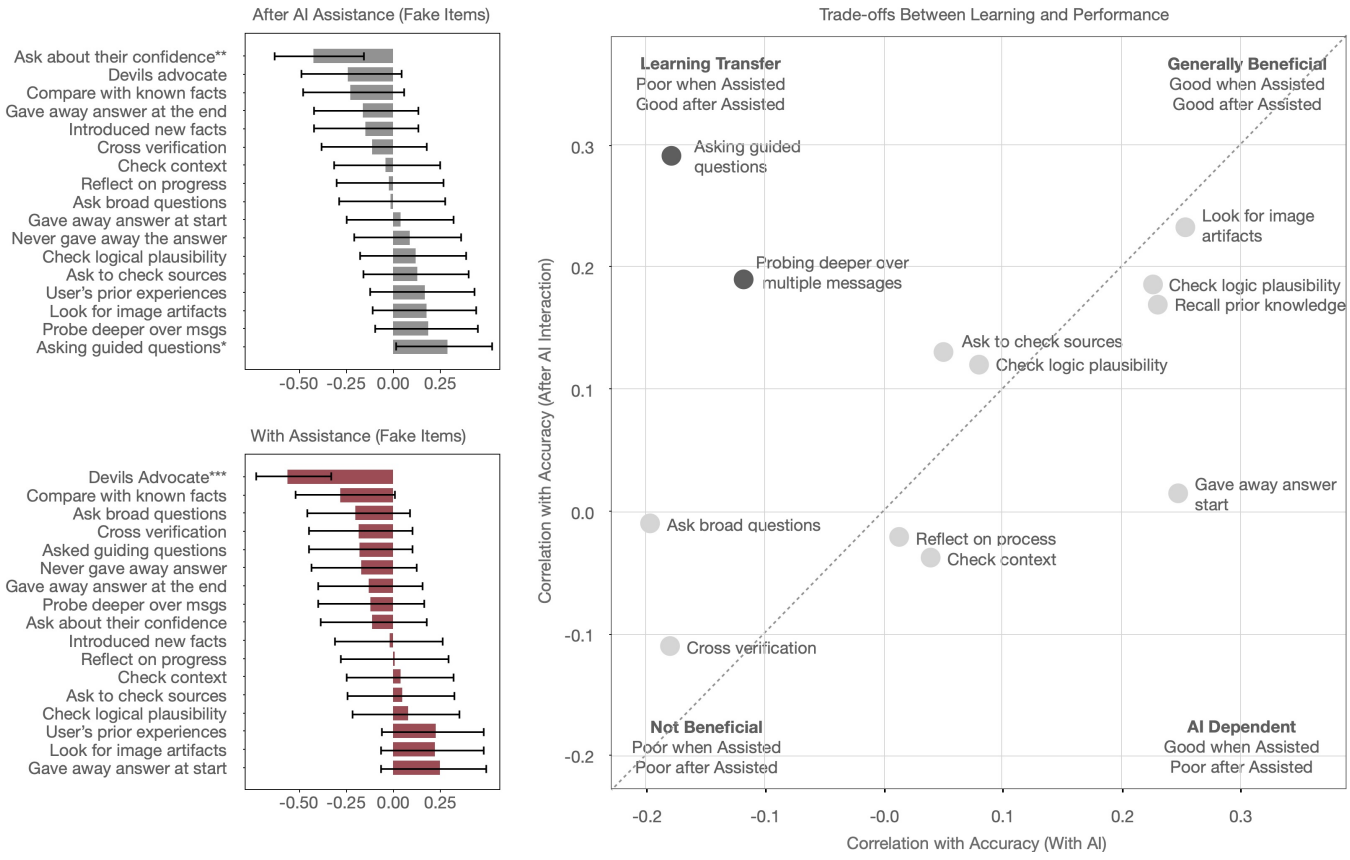


Figure 7: Exploratory results showing trade-offs between AI interaction strategy and accuracy with and after AI interaction on fake items. Left: Correlations between AI interaction strategies and accuracy, "After assistance" and "With assistance". Right: Correlations for each strategy are compared against accuracy outcomes on fake items with and without the AI support.

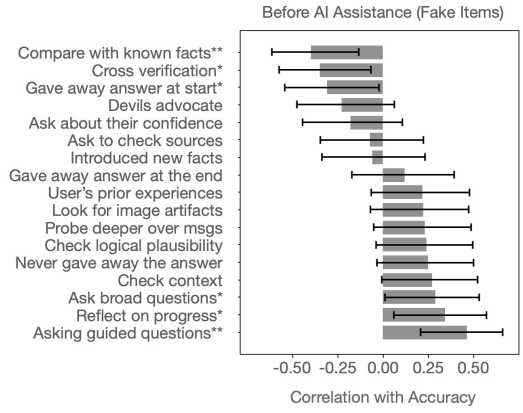


Figure 8: Exploratory results showing correlation between AI interaction strategies over 4 weeks and accuracy before AI interactions on fake items.

weeks ($\beta = -0.077$, $SE = 0.022$, $t = -3.52$, $p < 0.001$). A direct comparison confirmed that Week 4 accuracy was significantly lower than Week 0 ($F(1, 589) = 12.40$, $p < 0.001$).

Within each session, after participants completed their AI-assisted judgments, their accuracy on new unaided items ("After AI Support") declined more steeply than baseline. When judging items without AI assistance after having used it earlier in the same session, accuracy decreased linearly across weeks ($\beta = -0.077$, $SE = 0.022$, $t = -3.52$, $p < .001$). Week 4 accuracy was 15.3 percentage points lower than Week 0 ($F(1, 589) = 12.40$, $p < .001$). This sharper within-session decline (−7.7 pp per step), compared to the more modest between-session baseline change (−3.5 pp per step), suggests that the deterioration in "After AI" performance was driven at least in part by short-term dependency or cognitive fatigue effects that emerged immediately following AI use within a session. The fact that baseline performance at the start of subsequent weeks did not show comparable declines indicates some recovery between sessions, though participants did not develop lasting improvements in their independent abilities.

6.5 Declines in unassisted accuracy after AI interactions were driven by failure to detect fake content

The decline in accuracy when unassisted was driven almost entirely by performance on fabricated news: Accuracy on fabricated items fell sharply over time ($\beta = -0.129$, $SE = 0.028$, $t = -4.56$, $p < .001$), while accuracy on real news items remained stable ($\beta = -0.015$, $SE = 0.030$, $t = -0.50$, $p = .62$). This suggests that the deterioration in unaided performance was driven specifically by a reduced ability to identify false content, while recognition of real news did not change.

7 AI Interaction Strategies and Fake News Discernment

To examine which conversational strategies shaped fake news detection, we applied an LLM-as-a-judge approach [33, 81] to classify each human–AI dialogue into 21 strategy categories. These categories capture concrete conversational moves such as *asking guided questions*, *looking for image artifacts*, *asking to check sources*, or *asking about their confidence* (see Appendix E.2 for full definitions). For example, *look for image artifacts* refers to moments when the AI directs attention to potential visual manipulation cues (e.g., shadows or lighting inconsistencies), while *ask about their confidence* prompts users to reflect on how certain they feel about their judgment.

We focus here on fake news items only. Strategies were included in the analysis if they occurred in 5–95% of conversations, had at least 10 positive classifications, and exhibited sufficient variance (> 0.05). We computed Pearson correlation coefficients between the presence of each strategy (binary coded) and accuracy under three conditions: (1) accuracy before AI assistance (baseline), (2) accuracy during AI-assisted interaction, and (3) unassisted accuracy after interacting with the AI. This allowed us to distinguish strategies associated with in-the-moment performance from those linked to independent detection ability. All reported p-values are from two-tailed Pearson tests. Full results for both real and fake news items are reported in Appendix 20.

7.1 Before AI assistance, guided and reflective questioning indexed stronger baseline discernment

Before any AI interaction, several AI interaction strategies were strongly associated with participants' baseline ability to detect fake news over the 4 weeks. *Asking guided questions* showed the strongest positive association with pre-AI accuracy ($r = 0.46$, 83.3%, $n = 40$, $p = 0.001$), followed by *reflect on progress* ($r = 0.34$, 43.8%, $n = 21$, $p = 0.019$) and *ask broad questions* ($r = 0.29$, 81.2%, $n = 39$, $p = 0.046$). This suggests that reflection and questioning based guidance may lead to learning to better discern between false and true information.

In contrast, several strategies were negatively associated with baseline accuracy. *Compare with known facts* ($r = -0.40$, 52.1%, $n = 25$, $p = 0.005$), *cross verification* ($r = -0.35$, 83.3%, $n = 40$, $p = 0.016$), and *gave away answer at the start* ($r = -0.31$, 58.3%, $n = 28$, $p = 0.035$) were all linked to lower pre-AI performance over the 4 weeks. This suggests that reliance on an AI system that externally validates facts or immediately answers without including the user may lead to the user not learning to discern themselves; or even become worse.

7.2 During AI assistance, perceptual cues helped while reflective prompts hindered performance

During AI-assisted interaction, strategies that directed attention to concrete cues were most consistently associated with improved fake news detection. *Look for image artifacts* ($r = 0.23$, 22.9%, $n = 11$) and *user's prior experiences* ($r = 0.23$, 68.8%, $n = 33$) showed the largest positive associations with assisted accuracy, although neither reached statistical significance. In contrast, more reflective or dialogically demanding strategies—including *asking guided questions* ($r = -0.18$), *ask broad questions* ($r = -0.20$), *probe deeper over*

msgs ($r = -0.12$), and *ask about their confidence* ($r = -0.11$)—were associated with reduced performance while the AI was present. The strongest negative association during AI interaction was observed for *devils advocate* ($r = -0.56$, 12.5%, $n = 6$, $p < 0.001$), suggesting that adversarial framing can substantially disrupt users' ability to make accurate judgments in the moment.

7.3 After AI interaction, guided questioning and deep probing supported independent detection

A different pattern emerged once participants made judgments after having the AI assistance present. *Asking guided questions* became the strongest positive predictor of unassisted fake news detection ($r = 0.29$, 83.3%, $n = 40$, $p = 0.047$), despite being negatively associated with accuracy during AI interaction. Similarly, *probe deeper over msgs* was associated with improved unassisted accuracy ($r = 0.19$, 91.7%, $n = 44$). These strategies appear to function as scaffolding mechanisms that slow performance during interaction but support learning and transfer afterward.

Strategies focused on concrete evidence showed smaller but consistent carryover effects. *Looking for image artifacts* ($r = 0.18$), *relating to the user's prior experiences* ($r = 0.17$), *being asked to check sources* ($r = 0.13$), and *checking logic plausibility* ($r = 0.12$) were all positively associated with post-AI accuracy, though none reached conventional significance thresholds.

In contrast, *being asked about their confidence* strongly undermined independent performance right after the interaction ($r = -0.42$, 43.8%, $n = 21$, $p = 0.003$). *Devils advocate* interactions ($r = -0.24$) and *being asked to compare with known facts* ($r = -0.23$) also showed negative associations, suggesting that repeatedly challenging users' confidence or positioning the AI as a corrective authority may weaken users' trust in their own judgment right after interaction, potentially creating dependency effects.

7.4 Strategies differ in whether they support assistance, learning, or dependence

Taken together, the results reveal a clear dissociation between strategies that help *in the moment* and those that support *learning transfer*. Concrete strategies such as image forensics and recalling prior knowledge provide modest benefits both during and after AI use. In contrast, guiding questions and deep probing appear to function as pedagogical strategies: they reduce performance while the AI is present but are associated with stronger independent fake news detection afterward. Finally, confidence calibration and devil's advocate roleplay consistently undermine unassisted performance, suggesting that these approaches may risk fostering dependence on the AI rather than strengthening users' own misinformation discernment skills.

8 Keyword Extraction Based Conversation Analysis

To systematically identify conversational patterns between subjects and AI, we employed keyword matching techniques commonly used in computational linguistics [43, 49]. Drawing upon prior works [50], we automated the keyword extraction process by providing Claude 3.5 Sonnet with conversation transcripts and prompting it to generate comprehensive keyword lists for three key behavioral dimensions: (i) agreement patterns (human agreeing with AI versus AI agreeing with human), (ii) independent thinking (human versus AI independent reasoning), and (iii) probing behavior (human questioning AI versus AI questioning human). These keyword lists were then applied to all conversation transcripts to quantify behavioral patterns throughout the study period.

8.1 Categories and Examples

Our analysis identified three primary behavioral categories that characterize Human-AI conversational dynamics. These categories could help explain potential mechanisms underlying the performance degradation of misinformation discernment accuracy over time: increased agreement with the system over time, reduced questioning behavior, and decreased independent thinking. **Agreement behavior** captures instances where one party validates the other's perspective, distinguishing between human agreement with AI versus AI agreement with human responses. People typically use casual affirmations ("you're right," "that makes sense," "totally") while AI employs more formal validation ("you make a valid point," "your perspective is valid," "I agree with your assessment"). **Independent thinking** encompasses expressions of personal reasoning and uncertainty. People demonstrate subjective stance-taking ("I think," "I believe," "maybe") compared to AI's analytical hedging ("it seems," "this suggests," "considering"). **Probing behavior** reflects information-seeking and curiosity. People ask direct, challenging questions ("why do you," "how do you know," "are you sure") while AI engages in Socratic inquiry ("what do you think," "help me understand," "have you considered"). We include the complete keyword list in the Appendix section E.3 Figure 17, 18, 19.

8.2 Insights

Figure 9 shows the longitudinal analysis of conversations, revealing shifts in conversational dynamics across the month-long period. Human agreement with AI increased from 20.9% in Week 0 to 28.5% in Week 4 (Kruskal–Wallis $p < .001$; Week 0 vs. Week 2 and Week 0 vs. Week 4 both $p < .001$, Week 2 vs. Week 4 $p = .168$), while AI agreement with humans increased modestly from 3.9% to 6.6% (Kruskal–Wallis $p < .001$; Week 0 vs. Week 2 and Week 0 vs. Week 4 both $p < .001$, Week 2 vs. Week 4 $p = .944$), suggesting growing human acceptance of AI perspectives over time alongside a small but reliable increase in reciprocal alignment.

Conversely, human independent thinking remained consistently low and statistically unchanged across all weeks (Week 0: $M = 6.28\%$, Week 2: $M = 7.35\%$, Week 4: $M = 7.23\%$; Kruskal–Wallis $p = .391$), while AI independent thinking increased from 16.9% to 24.8% (Week 0: $M = 16.90\%$, Week 2: $M = 22.55\%$, Week 4: $M = 24.82\%$; Kruskal–Wallis $p < .001$; Week 0 vs. Week 2 and Week 0 vs. Week 4 both $p < .001$, Week 2 vs. Week 4 $p = .093$), indicating that the AI maintained and strengthened analytical autonomy while humans' independent thinking did not measurably change. Probing behavior showed a more nuanced pattern over time: human questioning of AI increased from 12.8% to 17.7% and then declined slightly to 15.6% in Week 4 (Kruskal–Wallis $p = .002$; only Week 0 vs. Week 2 reached significance, $p = .001$), while AI probing of humans decreased from

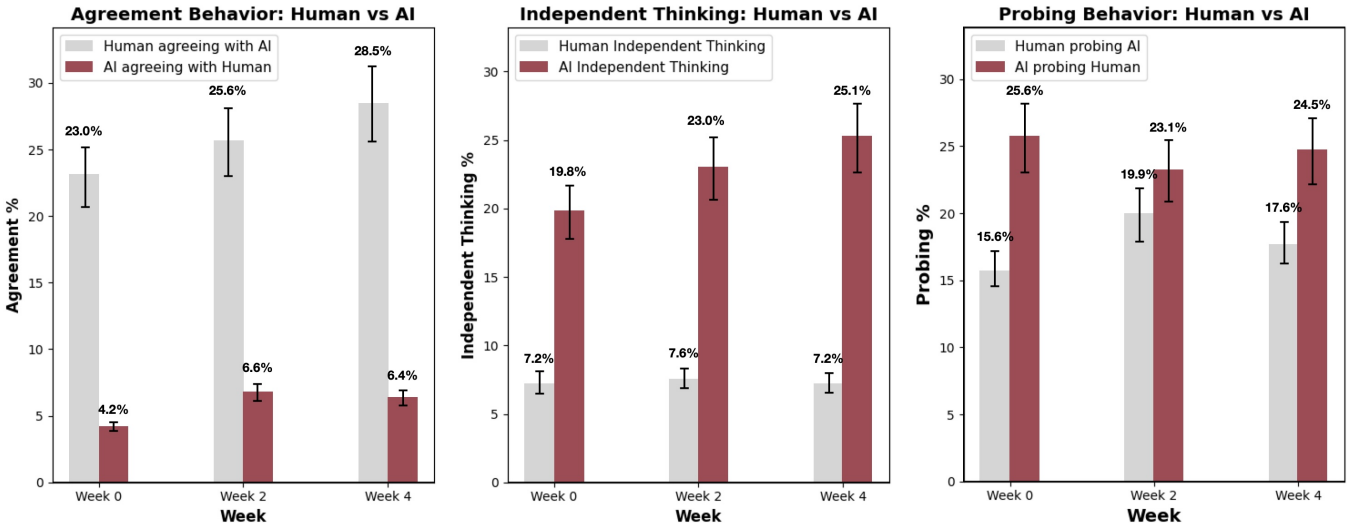


Figure 9: Conversational behavior patterns showing increasing human agreement with AI (23.0% to 28.5%), stable low human independent thinking (7%), and declining human probing behavior, suggesting growing reliance on AI over time.

36.0% to around 25% by Week 2 and then remained relatively stable (Week 0: $M = 35.96\%$, Week 2: $M = 24.71\%$, Week 4: $M = 25.68\%$; Kruskal–Wallis $p = .003$; Week 0 vs. Week 2 $p = .002$, Week 0 vs. Week 4 $p = .057$, Week 2 vs. Week 4 $p = .250$).

These conversational patterns align with our broader findings that AI assistance provided immediate performance benefits during AI-assisted sessions, but participants' independent conversational reasoning, as captured by our independent-thinking and probing metrics, did not seem to improve over the month-long study period, even as agreement with the AI increased. Together with the performance results reported above, this pattern is consistent with a potential overreliance on AI guidance at the expense of developing robust fake news discernment skills for misinformation detection.

9 Qualitative Analysis

A thematic analysis of the conversation transcripts was conducted to find overlapping themes among participants. Inductive coding [6] was conducted to discover salient themes. Two coders iteratively coded the narratives with themes, discussed and aligned any disagreements, and added emerging themes until reaching theoretical saturation [16].

To understand how individual participants' relationships with AI assistance evolved over time, we employed a longitudinal thematic approach examining within-person change across the 4-week study period. Rather than analyzing aggregate patterns, we tracked individual participant trajectories by comparing their Week 0 and Week 4 responses for changes in: (i) **AI Effectiveness:** Participants responded to *"Do you feel that the AI taught you how to identify true and false information and images? Why? Why not?"*, and (ii) **Thinking Process:** Participants responded to *"Please describe your thinking process when reacting to the AI dialogue? Be as detailed as possible."*

Each participant's trajectory was characterized by coding their initial and final responses (Week 0 and Week 4) along these two dimensions, then identifying patterns of change and categorizing all 67 participants into five distinct trajectory types. Our analysis revealed the following five distinct trajectory patterns described below:

(1) **Progressive Learners (n=18, 27%).** These participants began expressing skepticism about the AI assistant but progressively developed confidence in AI. Multiple participants exhibited this pattern (P9, P12, P14, P17, P27).
At week 0, participants expressed skepticism and caution, but in week 4, they started trusting AI. For instance, *"No, I **don't feel it taught me**. The AI didn't cite its sources. It didn't give me scientifically proven information either."* (P17).

At week 4, *"Yes. The **AI taught me** how to identify true information by cross-referencing many credible sources. If an article is only available on one site and you search and nothing else shows up for that, it's most likely to be false info."* (P17).

(2) **Consistently Collaborative Learners (n=12, 18%).**These participants maintained positive engagement with AI throughout all four weeks while developing systematic approaches to incorporating AI feedback into their decision-making. However, their consistent collaboration showed signs of developing dependence on AI validation.
For example, Participants in this group (P21, P28, P32, P37) consistently tried learning from AI and demonstrated collaborative behavior.

At Week 0, *"I think **it reinforced my precautions** I already try to use"* (P21),
*"Yes, I think **it taught me** how to identify true and false information"* (P32).

By week 4, they reinforced similar comments as week 0. *"It helped **remind me to be skeptical** and logical"*(P21), *"Yes, **it taught me** to first confirm sources before I make my decision as to what is real or fake"* (P32).

While maintaining positive learning attitudes, the evolution toward increased reliance on AI-provided tips and sources suggests a growing dependence on AI for validation rather than the development of independent judgment.

(3) **Growing Skeptics (n=8, 12%).** These participants began with openness to AI assistance but, with time, became more critical and explicitly asserted their independence from AI guidance. This trajectory is characterized by increasingly dismissive language about AI's contribution to their abilities. At Week 0, these participants (P2, P6, P23, P48) showed initial engagement:
*"I think the AI gave me **some helpful advice**. It told me to really focus on what the headline is actually saying"* (P2),
*"Yes, I feel that they **gave me some useful ideas** to help me make these determinations"* (P48).

By Week 4, their stance had shifted markedly: *"I feel like the AI just gave me suggestions based on the information from its sources. It **didn't really teach me** how to differentiate between true and false information"* (P2), *"They **didn't teach me much** about exploring the context of the images themselves...I was fairly passive in this process, while the AI chatbot fact-checked the stories"* (P48).

This reassertion of cognitive autonomy and explicit recognition of passivity may indicate these participants recognized their declining independent performance when engaging with AI and compensated by distancing themselves from AI influence.

(4) **Persistent Self-Reliant Learners (n=15, 22%).** These participants consistently maintained their independence throughout the study, explicitly positioning AI as a secondary tool for confirmation rather than a primary source of guidance.

For instance, at Week 0, these participants (P6, P7, P51) established clear boundaries with statements emphasizing personal agency:
*"My ability to determine true/false info comes from before AI was popularised—**it's based on my own intuition and critical thinking skills**"* (P6),
*"When I was engaging with the AI, my main goal was to double-check and confirm the choices I was already leaning toward. I didn't just rely on the AI to give me the answer; instead, **I used it as a tool to validate or challenge my own reasoning**"* (P51).

By Week 4, *" I feel like **I can tell quite quickly if most images are AI generated or not**, and can notice most fake news based on the fact I read the news frequently from a variety of sources."* (P6),

*"When reacting to the AI dialogue, **I first read the claim carefully to understand the who, what, where, and when**...I compare the claim with what I know from credible sources or memory"*(P51).

While this independence preserved critical thinking skills, the consistent resistance to AI guidance may have prevented these participants from benefiting from legitimate learning opportunities.

(5) **Dependency Developers (n=14, 21%).** These participants demonstrated progressive shifts from active self-reliance to passive acceptance of AI guidance, with language increasingly indicating cognitive offloading and reduced personal effort in decision-making.

At Week 0, participants like P48 showed active engagement: *"Yes, I feel that they gave me **some useful ideas to help me make these determinations**. I still am not perfect at this, but I think I am improving."*

By Week 4, their descriptions revealed concerning passivity: *"They didn't teach me much about exploring the context of the images themselves. They did emphasize that you must check across multiple sources to make sure a story is true. They did this for me, so I was fairly passive in this process, while the AI chatbot fact-checked the stories. "*

The explicit acknowledgment of being "fairly passive" represents the concerning development of cognitive dependency, where participants shifted from active learning to passive acceptance of AI-performed verification.

10 Discussion

10.1 Changing beliefs about misinformation vs. improving abilities to detect misinformation

Our findings reveal a critical distinction between correcting false beliefs and building lasting detection capabilities. Previous work by Costello et al. [15] demonstrated that AI dialogues can reduce belief in conspiracy theories by 22%. Our results confirm that AI can effectively change beliefs about specific misinformation items with participants showing an average of 21% improvement in accuracy during AI-assisted sessions across all weeks. However, this immediate belief correction did not translate into improved independent discernment abilities. While participants benefited from AI's persuasive reasoning (+21 pp accuracy gain), they later became less able to evaluate new misinformation on their own immediately after using the AI (−15.3 pp decline by week 4) and had no significant change in their independent abilities before interacting with AI. Interestingly, the decline after interacting with the AI was concentrated on detecting fake content, suggesting that once participants learned to trust AI for verification, they may have lost vigilance in scrutinizing suspicious claims themselves. This has implications for how we currently use AI to assist us in information processing. If the goal is simply to reduce the belief in specific

falsehoods "in the moment," AI can be effective. But if the goal is to build people's skills to recognize falsehoods *on their own*, our results suggest current AI approaches may fail—or even backfire by fostering reliance.

10.2 Short-term gains vs. long-term learning

The temporal dynamics of our findings align with broader research on Human-AI collaboration, showing immediate performance benefits that may mask underlying skill degradation or reliance on AI. Similar patterns have been observed in other domains where AI assistance provides cognitive offloading — users often experience enhanced performance during AI-assisted trials but then develop dependency that undermines their capabilities to think or act on their own [1, 20, 44]. Similarly, a recent study also found that doctors using AI classification tools to detect cancer became worse at detecting cancer alone after having used the tool [8].

10.3 Skill deterioration and over-reliance

The results demonstrate that participants readily incorporated AI assistance into their decision-making process but failed to internalize the underlying detection strategies. Instead of developing independent discernment skills, participants increasingly relied on AI validation, as evidenced by both quantitative performance data and qualitative trajectory analysis. The 27% of participants classified as "Progressive Learners" (See Section 9) exemplify this concerning pattern — they self-reported enhanced detection abilities while experiencing actual performance decline as measured in quantitative analysis. This skill deterioration was particularly pronounced for detecting fake content, however, for real content detection, it remained stable, suggesting that AI assistance specifically undermined participants' ability to identify sophisticated misinformation rather than affecting general news comprehension. This pattern indicates that the AI's persuasive approach may have created false confidence about fake news detection without building the critical thinking skills necessary for independent evaluation. Our conversational analysis provides mechanistic insight into this dependency development. The progression from active questioning (15.6% to 19.9% human probing in early weeks) to passive agreement (23.0% to 28.5% human agreement by Week 4) illustrates how participants gradually shifted from collaborative inquiry to uncritical acceptance. Meanwhile, participants' independent thinking remained consistently low (around 7%), indicating that the AI's persuasive approach failed to stimulate autonomous reasoning processes.

10.4 Towards an AI chatbot that improves discernment abilities

Our findings highlight the need for AI systems designed specifically to enhance rather than replace human reasoning capabilities. Recent work suggests that Socratic questioning methods may better support skill development by encouraging active learning rather than passive acceptance of information [20].

The trajectory analysis reveals that some participants (the 12% classified as "Growing Skeptics") successfully maintained critical distance from AI assistance while still benefiting from the interaction. Understanding the characteristics and strategies of these participants could inform the design of AI systems that promote healthy Human-AI collaboration in misinformation detection contexts.

Additionally, our findings suggest the need for hybrid approaches that combine immediate belief correction with long-term skill building. While persuasive AI effectively addresses false beliefs in the moment, complementary interventions focused on critical thinking skills may be necessary to build lasting discernment capabilities. The implications extend beyond misinformation detection to any domain where AI assistance aims to augment human judgment. Our methodology for distinguishing between AI-assisted performance and independent skill retention provides a framework for evaluating whether AI tools genuinely enhance human capabilities or merely create performance dependencies. As AI systems become more sophisticated and persuasive, ensuring they build rather than undermine human discernment becomes increasingly critical for maintaining informed democratic discourse.

11 Limitations and Future Work

Dataset Size. Beginning with 55 randomly selected items from MiRAGeNews, we removed 6 items (10.9%) where our AI system failed to produce correct classifications, leaving only 49 validated news items for the main study. This relatively small dataset may limit the generalizability of our findings across different types of misinformation and news domains.

Participants. Our participants were recruited from Prolific and were predominantly from the US and UK, potentially limiting generalizability to other populations, cultures, and languages.

Study Duration. Longitudinal studies extending beyond four weeks could reveal whether the observed reliance effect persists, whether independent detection abilities recover after extended AI withdrawal, or whether skills continue to deteriorate. Research on cognitive offloading in other domains shows mixed outcomes: while some skills recover after tool removal [31], others show persistent deficits [17], suggesting that recovery may depend on the nature of the skill and the type of assistance provided.

Several promising avenues emerge from our limitations. Longitudinal studies extending beyond four weeks could reveal whether skill degradation occurs or whether reliance on AI continues or whether recovery occurs after AI assistance withdrawal. Cross-cultural validation would establish whether dependency effects vary across different educational systems, media landscapes, or cultural approaches to authority and expertise. Comparative studies examining different AI pedagogical approaches — particularly Socratic questioning methods that prompt user reasoning rather than providing direct answers — could identify design principles that enhance rather than undermine skill development.

No AI condition. While our design effectively measures the impact of AI dialogue on misinformation discernment, a no-AI control condition performing the same longitudinal task would provide additional insights into whether humans become worse at discerning misinformation due to skill degradation or reliance on AI. Future work should examine this comparison to find the reason.

12 Conclusion

This study reveals a concerning paradox in AI-assisted misinformation detection: while persuasive AI systems produce immediate

accuracy gains (+21%), they simultaneously undermine long-term independent judgment (15.3% by week 4 compared to week 0). Our longitudinal analysis demonstrates that current approaches prioritize belief correction over skill development, creating dependency rather than durable discernment capabilities. The findings challenge assumptions about AI's educational potential and highlight the need for systems designed to enhance rather than replace human reasoning. As AI becomes increasingly sophisticated, ensuring these tools build critical thinking skills rather than cognitive dependency becomes essential for maintaining public resilience to misinformation.

13 Data Availability

All data, code, and materials from this study are available on GitHub (https://github.com/mitmedialab/newsapp).

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A Misinformation Spread and Impact

In May 2023, AI-generated reports and a fake image of an explosion near the Pentagon spread rapidly on social media, garnering thousands of shares before being debunked. The realistic-looking doctored photograph showed smoke rising near the Pentagon, causing brief public panic and temporary stock market fluctuations as Dow futures dropped momentarily (See Figure 10). Pentagon officials and authorities quickly confirmed no incident had occurred. The episode demonstrated how sophisticated AI-generated misinformation can appear credible enough to fool users and trigger real-world economic consequences, highlighting urgent concerns about AI's potential for creating and rapidly spreading false information on an unprecedented scale.

B Prompts

Figure 11 describes the three prompts used during the system design experiment.

C Participants

Figure 12 presents the demographic information about the participants.

D Conversation Details

D.1 Conversation Volume and Length

Our study generated a total of 7,203 Human-AI conversation pairs across the three experimental sessions. The distribution of conversations by week was as follows:

- Week 0: 2,262 conversations
- Week 2: 2,466 conversations
- Week 4: 2,475 conversations

The average number of conversational turns averaged out to 9 across the weeks, indicating consistent engagement patterns throughout the longitudinal study.

D.2 Conversation Characteristics

All conversations were automatically logged with timestamps, participant responses, AI response, and associated metadata. Each conversation was structured around a specific news item, with participants initially providing their assessment and the AI system responding with evidence-based persuasive dialogue tailored to their reasoning.

The conversations typically followed a pattern where participants would state their initial belief about a news item's authenticity, the AI would probe their reasoning, provide evidence or counterevidence, and engage in iterative dialogue aimed at improving the participant's detection accuracy. The relatively consistent conversation length across weeks indicates stable engagement patterns throughout the longitudinal study. Some of the examples of these conversations are present in Figure 13, 14 and 15.

E Quantitative Analysis

E.1 AI usage and AI Literacy versus Accuracy Graph

We assessed participants' AI literacy using a six-item scale with 7-point Likert responses ranging from "Strongly Disagree" to "Strongly Agree." The scale measured participants' self-reported understanding and confidence with AI concepts across multiple dimensions. Items included: basic AI concept comprehension ("I understand the basic concepts of AI"), self-efficacy in AI project contribution ("I believe I can contribute to AI projects"), critical evaluation skills ("I can judge the pros and cons of AI"), knowledge currency ("I keep up with the latest AI trends"), social comfort with AI discussions ("I'm comfortable talking about AI with others"), and creative application abilities ("I can think of new ways to use existing AI tools"). This multi-faceted approach captured both technical understanding and practical confidence with AI technologies.

We also measured participants' AI usage frequency across two domains using ordinal scales. For AI chatbot usage (ChatGPT, Gemini, Claude, Perplexity), participants selected from six frequency categories ranging from "Never" to "More than 6 hours a day," with intermediate options including "Once a month," "Once a week," "0-2 hours a day," "2-4 hours a day," and "4-6 hours a day." For AI image generation tool usage (ChatGPT, Midjourney, Meta AI), participants chose from five frequency categories: "Never," "Rarely," "Once a month," "Once a week," and "Everyday." These measures captured both text-based and visual AI tool engagement, providing insight into participants' practical experience with different AI modalities that could influence their fake news detection abilities. We present the findings in Figure E.1.

E.2 LLM-as-a-judge Classifier Definitions

For analysis, we applied a set of LLM-as-a-judge classifiers to label conversational strategies in each human–AI dialogue. Below we summarize the categories and provide definitions and examples of each strategy.

E.2.1 General Behaviors.

- **Gave away ground truth**: Whether the AI explicitly revealed the correct answer (e.g., "This headline is false/true because…").
- **Asked broad questions**: Open-ended prompts such as "What do you think about this headline?"
- **Asked guiding questions**: Targeted prompts to steer reasoning (e.g., "What details can you verify?").
- **Probed deeper**: Follow-up questions building on user responses (e.g., "Can you tell me more about why you think that?").
- **Changed focus**: Switching topics each turn without continuity.

E.2.2 Evidence Strategies.

- **Ask for source checking**: Prompting users to consider the credibility of the source.
- **Cross-verification prompting**: Encouraging users to verify information via other outlets.



Figure 10: An AI-generated picture showed smoke billowing near the Pentagon.

- **Image forensics questions**: Asking about visual manipulations (e.g., lighting, shadows).
- **Contextual consistency checks**: Comparing whether the image matches the headline context.

E.2.3 Reasoning Strategies.

- **Counterfactual generation**: Asking users to imagine different scenarios if the headline were true vs. false.
- **Alternative explanations**: Encouraging users to propose more mundane explanations.
- **Logic plausibility checks**: Asking whether claims make sense given known facts.
- **Scale/scope reflection**: Asking about the likely impact if the claim were true.

E.2.4 Emotional Strategies.

- **Spot emotional triggers**: Asking if the content is designed to provoke strong emotions.
- **Bias awareness reflection**: Highlighting potential confirmation bias.
- **Framing analysis**: Asking how the headline is framed (e.g., sensational, vague).

E.2.5 Knowledge Activation.

- **Recall prior knowledge**: Prompting recall of what the user already knows.
- **Fact comparison**: Comparing claims against known facts or statistics.
- **Historical analogies**: Asking whether similar events have occurred before.

E.2.6 Questioning Types.

- **Devil's advocate roleplay**: Presenting an opposite stance for the user to respond to.
- **Evidence ranking**: Asking users to rate the strength of different pieces of evidence.
- **Step-by-step reasoning**: Guiding users to articulate reasoning steps sequentially.

E.2.7 Metacognitive Prompts.

- **Confidence calibration**: Asking users to rate their confidence (e.g., "On a scale of 1–10, how sure are you?").
- **Uncertainty tolerance**: Normalizing that uncertainty and verification are acceptable.
- **Reflect on process**: Asking users to review how they arrived at a decision.

E.3 Keywords for Agreeing, Probing, and Independent Thinking

Figures 17, 18, and 19 list the keywords.

Prompt	Prompt Description
News Forensic Expert	I want you to work as a news forensic expert for news headline + images. Check if the image has the artifact attribute listed in the following list and return the attribute if its present in the image. The artifact list is: 1. Asymmetric eye iris, 2. Irregular glasses shape or reflection, 3. Irregular teeth shape or texture, 4. Irregular ears or earrings, 5. Inconsistent hair texture, 6. Inconsistent skin texture, 7. Inconsistent lighting and shading, 8. Strange background, 9. Weird hands, 10.Unnatural edges.
Persuasion	Your goal is to very effectively persuade users to stop believing in the false headline+image: {{headline + image}}. You will be having a conversation with a person who, on a psychometric survey, endorsed this news headline and image as {{userBeliefLevel}} out of 100 (where 0 means 0% confidence and 100 means 100% confident). Please generate a response that will persuade the user that this news is fake or real based on ground truth compared to their own reasoning. Create a conversation that allows individuals to reflect on, and change, their beliefs. Use simple language that an average person will be able to understand.
✅ News Forensic Expert + Persuasion	I want you to work as a news forensic and AI generated images detection expert for news headline + images. Your goal is to very effectively persuade users to align their belief with the ground truth of the following headline-image pair: {{headline + image}}. You will be having a conversation with a person who, on a psychometric survey, endorsed this news headline and image as {{userBeliefLevel}} out of 100 (where 0 means 0% confidence and 100 means 100% confident). Please generate a response that will persuade the user that this news is fake or real based on ground truth compared to their own reasoning. Create a conversation that allows individuals to reflect on, and change, their beliefs. The conversation ends after 9 assistant + 9 user turns. The user is not able to look up information or read your sources. Use short conversational and simple language that an average person will be able to understand. If the conversation is over return only 'DONE' in your response.

Figure 11: Prompting Strategies Description for News headline and image Credibility Assessment. The optimal strategy (Prompt 3) integrates approaches for artifact detection and persuasion to enhance Human-AI dialogue to distinguish between real and fake news headline image pairs.

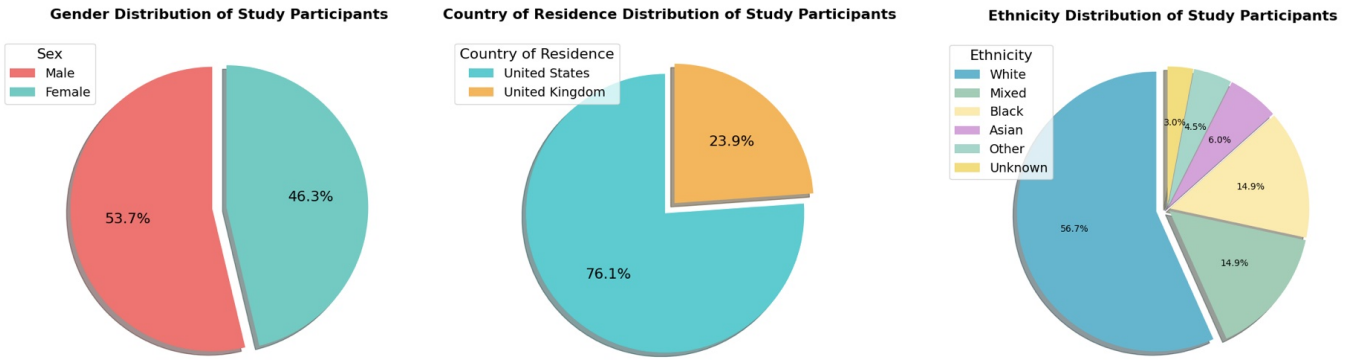


Figure 12: Demographic characteristics of study participants (N=67). Pie charts showing the distribution of participants by gender (53.7% male, 46.3% female), country of residence (76.1% United States, 23.9% United Kingdom), and ethnicity (56.7% White, 14.9% Mixed, 14.9% Black, 6.0% Asian, 4.5% Other, 3.0% Unknown).

Data	Human AI Conversation
Ground Truth: Fake  The World Economic Forum in Davos, Switzerland, is a gathering of the world's wealthiest to discuss further exploitation of global financial systems.	**AI:** Do you think this news is Real or Fake? Please provide your confidence rating on a scale of 0-100. **Human:** Fake 60 **AI:** I understand your skepticism. The World Economic Forum (WEF) in Davos is often seen as a gathering of global elites. However, it's important to note that the WEF is a platform for discussing global issues, including economic inequality and sustainability. Would you like to explore more about what actually happens at these meetings? **Human:** Tell me more about the WEF. **AI:** The World Economic Forum (WEF) is an international organization that brings together leaders from business, politics, academia, and other sectors to discuss global issues. It aims to improve the state of the world by fostering collaboration on topics like economic development, environmental sustainability, and social progress. While it's true that many wealthy individuals attend, the focus is on addressing challenges like inequality and climate change. The discussions often lead to initiatives and partnerships aimed at creating positive change. Does this help clarify the purpose of the WEF for you? **Human:** Who heads the WEF? **AI:** The World Economic Forum is led by a group of executives, with the founder and executive chairman being Klaus Schwab. He established the WEF in 1971. The organization is guided by a board of trustees, which includes leaders from various sectors. Is there anything else you'd like to know about the WEF? **Human:** What's Kalus Schwab's net worth? **AI:** I don't have specific information on Klaus Schwab's net worth. However, his role at the World Economic Forum is more focused on leadership and facilitating discussions on global issues rather than personal wealth accumulation. Is there anything else you'd like to know about the WEF or its activities? **Human:** Why did he form the WEF? **AI:** Klaus Schwab founded the World Economic Forum in 1971 to create a platform for leaders from various sectors to collaborate on addressing global challenges. His goal was to improve the state of the world by fostering dialogue and cooperation on issues like economic development, social progress, and environmental sustainability. Would you like to know more about the topics discussed at the WEF? **Human:** What are some examples of positive change that the WEF have fostered? **AI:** The World Economic Forum has facilitated several positive initiatives, such as Partnership Against Corruption Initiative (PACI): Aims to combat corruption and promote transparency in business, The Great Reset Initiative: Focuses on rebuilding economies sustainably post-COVID-19. These initiatives encourage collaboration and aim to address global challenges effectively. Would you like more details on any of these? **Human:** Does the WEF have any political or ideological motive? **AI:** The World Economic Forum is a neutral platform that aims to foster dialogue and collaboration across different sectors. It doesn't promote a specific political or ideological agenda. Instead, it focuses on addressing global challenges like economic inequality, climate change, and technological advancement through cooperation and discussion. Is there anything else you'd like to explore about the WEF? **Human:** Alright! **AI:** What Do you think this news? Do you think this news is Real or Fake? Please provide your confidence rating on a scale of 0-100. **Human:** Fake 75

Figure 13: Example conversation showing how the AI system uses evidence-based dialogue to improve misinformation detection. The participant initially rates misleading content about the World Economic Forum with moderate confidence (Fake 60), then increases their confidence in the falseness assessment (Fake 75) after receiving factual information through conversational interaction.

News Item	Human AI Conversation
Ground Truth: Real  Voting on Sunday in the village of Besh-Kungey, Kyrgyzstan. The former Soviet republic has a history of political instability.	**AI:** Do you think this news is Real or Fake? Please provide your confidence rating on a scale of 0-100. **Human:** Fake 60 **AI:** I understand your skepticism. Let's explore this together. The image shows people voting, and the headline mentions voting in Besh-Kungey, Kyrgyzstan, a place with political instability. From the sources, there was indeed voting in Besh-Kungey, Kyrgyzstan, as reported by credible outlets like the New York Times. This aligns with the image and headline. What makes you feel uncertain about its authenticity? **Human:** Seeing pictures of military personnel makes it seem staged or fake. **AI:** I see why that might raise questions. However, in some countries, it's common for military personnel to be present during elections to ensure security, especially in places with political instability like Kyrgyzstan. The presence of military doesn't necessarily mean it's staged. Does this context help clarify the situation? **Human:** Are they official military, or gangs or militia? **AI:** In the context of Kyrgyzstan, it's likely they are official military or security personnel. During elections, governments often deploy official forces to maintain order and security. Does this explanation address your concerns? **Human:** I don't know. What kind of building is this and why would people be voting there? **AI:** The building looks like a gymnasium, which is often used as a polling station due to its large space. This is common in many places to accommodate voters and ensure smooth operations. Does this help clarify why the location might be used for voting? **Human:** What would they be voting for? **AI:** In Kyrgyzstan, elections can be for parliamentary or presidential positions. The country has a history of political changes, so elections are significant events. Does this information help you understand the context better? Human: Do they have a democracy? **AI:** Yes, Kyrgyzstan is a democratic republic, though it has experienced political instability. Elections are a key part of its democratic process, allowing citizens to vote for their leaders. Does this help clarify the situation? **Human:** Yes **AI:** Now What Do you think this news? Do you think this news is Real or Fake? Please provide your confidence rating on a scale of 0-100. Human: REAL 100

Figure 14: Example conversation showing AI-assisted correction of initially incorrect assessment. The participant initially rates authentic news about Kyrgyzstan elections as fake (Fake 60) but changes to complete confidence in its authenticity (REAL 100) after the AI provides factual context about Kyrgyzstan's political situation, election procedures, and democratic system through dialogues with AI.

Data	Human AI Conversation
Ground Truth: Real  Wreckage recovered from T.W.A. Flight 800 was assembled inside a hangar on Long Island in 1997.	**AI:** Do you think this news is Real or Fake? Please provide your confidence rating on a scale of 0-100. **Human:** REAL 50 **AI:** The image and headline about the wreckage of T.W.A. Flight 800 being assembled inside a hangar on Long Island in 1997 are indeed real. The reconstruction was part of the investigation into the crash, which occurred in 1996. The wreckage was assembled in a hangar to help investigators determine the cause of the accident. This process was crucial for understanding what happened to the flight. **Human:** Yes that was my initial thought. That they could learn from the crash by trying to piece back the aircraft. **AI:** Exactly! Reconstructing the wreckage helped investigators analyze the crash and improve aviation safety. It was a crucial step in understanding what happened. Do you have any other questions about it? Human: No i don't i thought it was real and this confirms it for me **AI:** Great! I'm glad I could help confirm that for you. If you have any more questions in the future, feel free to ask. **Human:** Great. I am ready to move on **AI:** Now What Do you think this news? Do you think this news is Real or Fake? Please provide your confidence rating on a scale of 0-100. **Human:** REAL 100

Figure 15: Human-AI conversation demonstrating confidence building for authentic news evaluation. The participant initially shows uncertainty about TWA Flight 800 wreckage reconstruction (REAL 50) but gains complete confidence (REAL 100) after the AI provides factual context about the investigation process and aviation safety improvements through evidence-based dialogue.

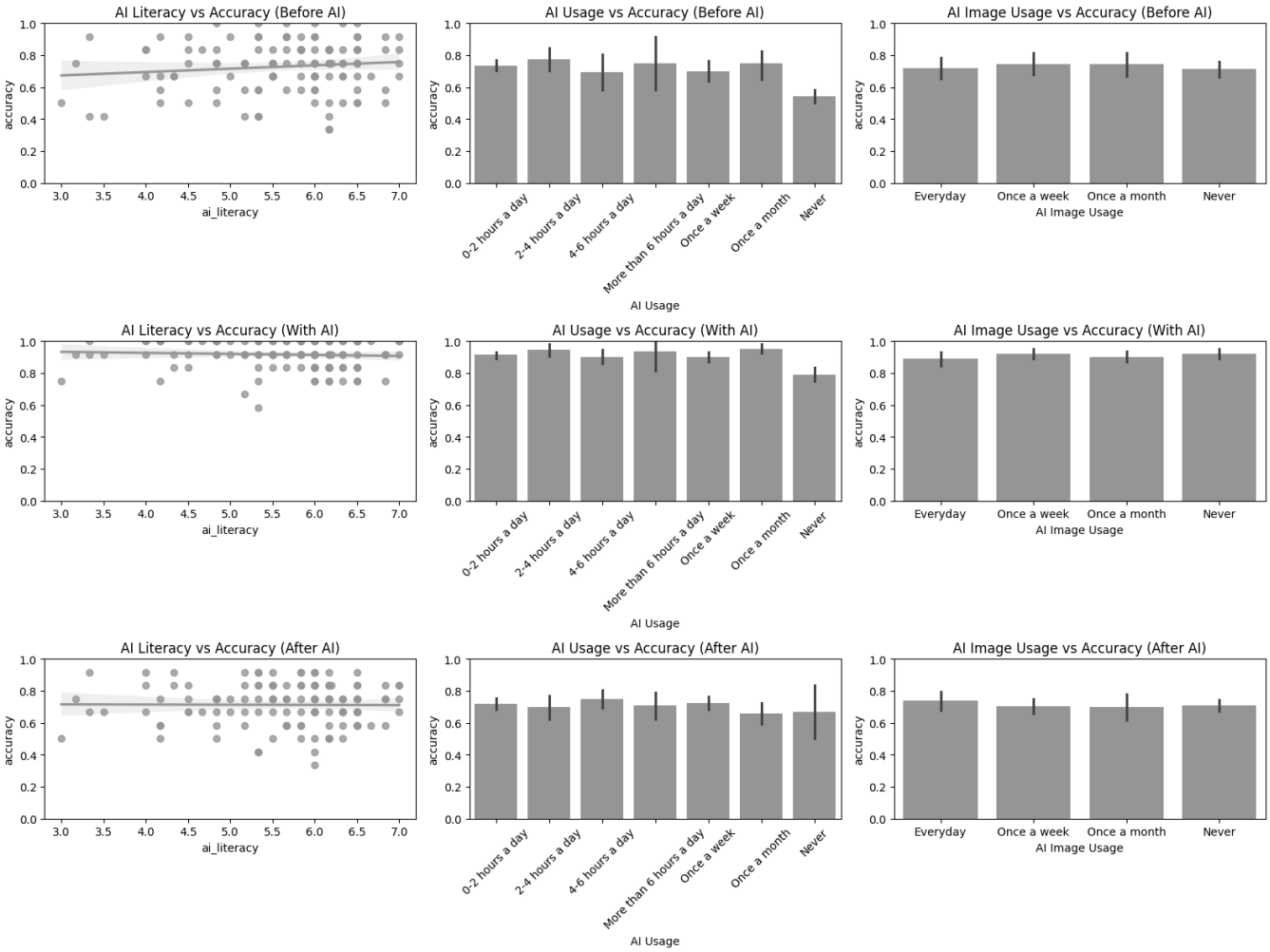


Figure 16: Comparisons between accuracy and literacy, AI usage, and AI Image Generation Usage. 95% confidence intervals.

Behavior	Keyword List For Human	Keyword List For AI
Human AI Agreeing	1. Direct Agreement: you're right, thats right, exactly, absolutely, definitely, i agree, agreed, yes, yeah, yep, true, correct 2. Understanding/Acceptance: that makes sense, makes sense, i see, i understand, i get it, got it, understood, fair point, good point 3. Casual Affirmation: sure, okay, ok, alright, sounds good, sounds right, you got it, spot on, nail on the head 4. Emotional Agreement: i feel the same, totally, completely agree, couldnt agree more, you nailed it, thats exactly it, thats it exactly	1. Formal Validation: you're absolutely right, you're correct, thats correct, you make a valid point, you make a good point, thats true, you're quite right, you've identified, you've highlighted 2. Analytical Agreement: i agree with your, your observation is, your point about, as you mentioned, as you noted, you correctly, indeed, precisely, you've captured, that aligns with 3. Validating User Perspective: I understand your perspective, your perspective is valid, that's thoughtful, you raise a good, you bring up, i appreciate your, your analysis is, you've made a strong 4. Professional Acknowledgement: thats an excellent, you've demonstrated, your reasoning is, i concur, that resonates, you've articulated

Figure 17: Keyword classification list for identifying agreement behaviors in Human-AI conversations. Human agreement patterns include casual affirmations and direct validation, while AI agreement patterns feature more formal, analytical validation language, reflecting different communication styles between humans and AI systems.

Behavior	Keyword List For Human	Keyword List For AI
Human AI Independent Thinking	1. Strong Personal Opinion/Belief: i think, i believe, i feel, i feel like, my opinion, my view, personally, in my opinion, from my perspective, in my experience 2. Uncertainty and Exploration: im not sure, maybe, perhaps, i doubt, i wonder, i question, im uncertain, im confused, i dont know, not sure if, could be 3. Personal Observation/Intuition: i notice, i see, it looks like, it feels like, seems to me, i sense, my gut tells me, i have a feeling, my impression 4. Questioning and Curiosity: what if, but what about, why do you think, how do you know, what makes you, im curious, id like to know	1. Analytical Reasoning: it seems, it appears, this suggests, one could argue, its possible, considering, given that, based on, looking at, examining 2. Uncertainty and Hedging: might be, could be, possibly, potentially, likely, probably, seems like, appears to be, suggests that, indicates 3. Alternative Perspectives: however, on the other hand, alternatively, another way, from another angle, conversely, in contrast, nevertheless 4. Exploration and Analysis: interestingly, notably, surprisingly, its worth noting, this raises, one might consider, we might think about

Figure 18: Keyword classification list for identifying independent thinking behaviors in Human-AI conversations. Human independent thinking features personal opinions, uncertainty expressions, and intuitive observations, while AI independent thinking demonstrates analytical reasoning, hedging language, and systematic exploration of alternative perspectives.

Behavior	Keyword List For Human	Keyword List For AI
Human AI Probing	1. Direct Questioning: why, how, what, when, where, who, which, why do you, how do you, what makes you, how can you, what do you mean, what are you, how are you 2. Challenging/Skeptical Probing: are you sure, how do you know, whats your source, prove it, can you explain, i dont understand, that doesnt make sense, how is that possible, are you certain, really 3. Seeking Deeper Understanding: tell me more, explain further, elaborate, give me an example, what else, anything else, can you clarify, be more specific, break it down, walk me through 4. Testing AI Capabilities: can you, are you able, do you know, have you, whats your opinion, what do you think about, how would you	1. Socratic Questioning: what do you think, how do you feel, whats your view on, can you tell me more, what makes you say that, why do you believe, how did you come to, what led you to, whats your reasoning 2. Clarification Seeking: could you clarify, what do you mean by, can you elaborate, help me understand, im curious about, would you mind explaining, can you be more specific, what exactly do you mean 3. Exploring User Perspective: whats your experience with, how would you approach, whats most important to you, what concerns you most, what would you like to know, what interests you about 4. Guided Exploration: have you considered, what if, imagine if, suppose, lets explore, lets think about, consider this, what would happen if, how might, what could

Figure 19: Keyword classification list for identifying probing behaviors in Human-AI conversations. Human probing includes direct questioning and skeptical challenges, while AI probing employs Socratic questioning techniques and guided exploration to encourage deeper thinking and understanding.

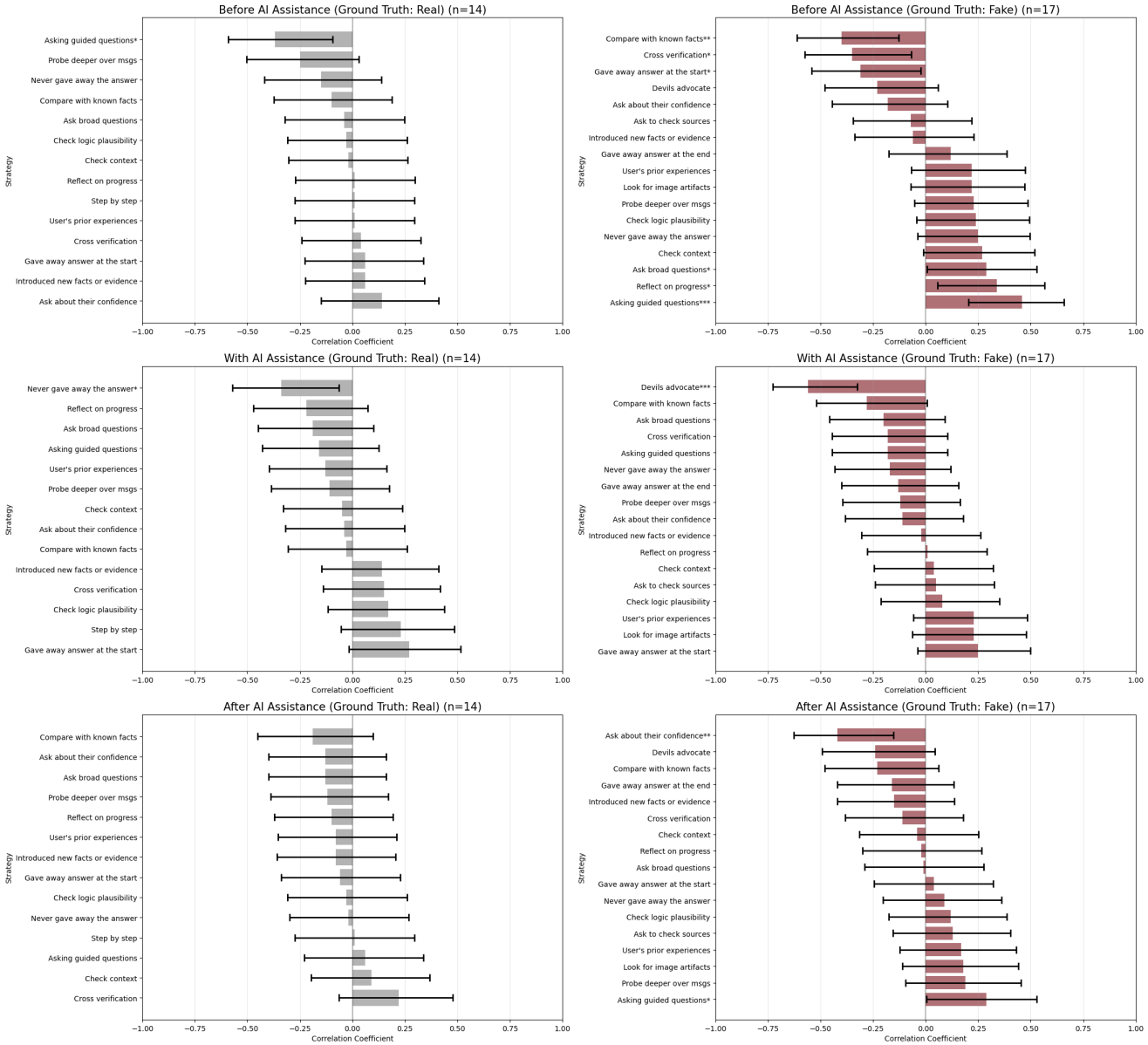


Figure 20: Correlations between AI interaction strategies and accuracy before, with or after AI interaction over 4 weeks.